DS608 GROUP PROJECT

Who Feels Inflation Most?

Decomposing Singapore's CPI by category and household income, 2019 to 2026

Jiang Liu Ru · Amsyar Minhaj · Su Liang-Yu

Live dashboard: ds608-sg-inflation.streamlit.app

THE PROBLEM

One headline rate hides who pays

Inflation hit **7.49%** in September 2022, the highest in over a decade, then cooled. Households still feel prices are high.

A single number hides two things: **which categories** pushed prices up, and whether the same price change hits **lower-income households** harder.

RESEARCH QUESTION

Which spending categories have driven Singapore's inflation since 2019, and have lower-, middle- and higher-income households experienced it differently?

DATA

Five public datasets, all aggregate

Dataset	Frequency	Used for
CPI, 2024 base (data.gov.sg)	Monthly, 1961–Aug 2026	Headline and 10 category indexes
CPI by household income group	Half-yearly	Lowest 20%, middle 60%, highest 20%
CPI weights (rebasing paper)	2019 and 2024 baskets	Category contributions
COE bidding results (LTA)	Monthly	Private transport link
HDB rents (from Jan 2021)	Monthly	Accommodation link

The 2024-base CPI is already officially linked back to 1961, so no base-year splicing was needed. All files are public statistics with no personal data.

Sources: SingStat, data.gov.sg (Singapore Open Data Licence), LTA, HDB · downloaded 24 Oct 2026 · 3

METHOD

Split headline inflation into category shares

$$\text{Contribution} = \text{weight} \times (\text{index now} - \text{index a year ago}) \div \text{all-items index a year ago} \times 100$$

The ten category contributions add up to headline year-on-year inflation, in percentage points

1 · Wrangle

Reshape wide SingStat tables to long format, label each series by level

2 · Decompose

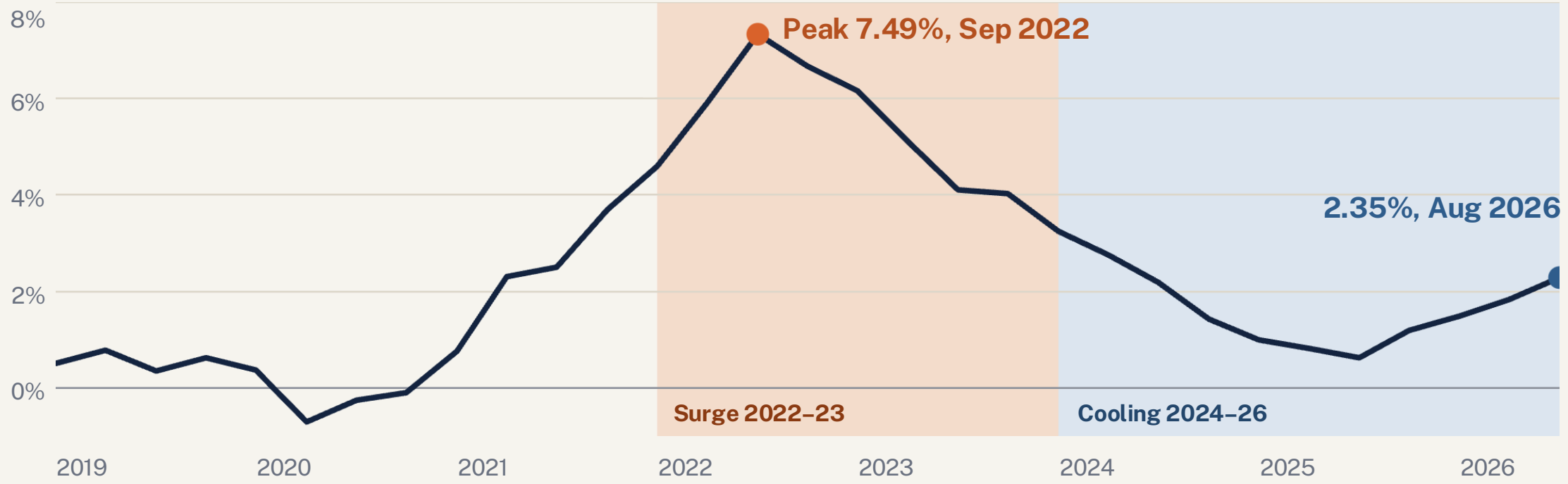
2019 weights for 2019–2023, 2024 weights from 2025, on matching index bases

3 · Compare

Income groups by half-yearly inflation rates and percentage-point gaps

FINDING 1 · HEADLINE

Inflation surged in 2022, then cooled

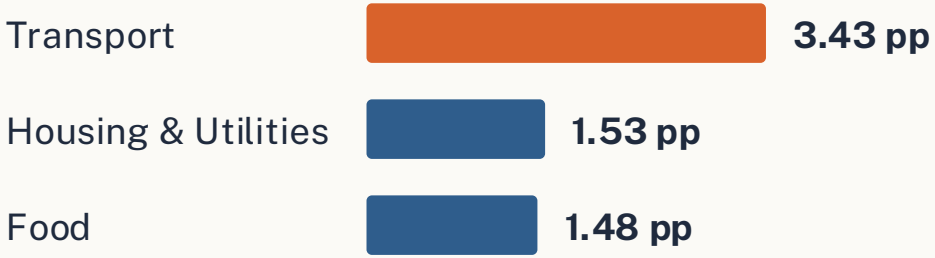


Headline CPI, year-on-year %, quarterly average of monthly rates · SingStat · 5

FINDING 2 · TOP CONTRIBUTORS

Transport, Food and Housing drive it

Peak month: Sep 2022 (7.49%)



2019 basket weights

Latest month: Aug 2026 (2.35%)



2024 basket weights

Transport is the largest contributor in both months. The basket changed in 2024, so we compare rankings across periods, not sizes.

Contribution to headline year-on-year inflation, percentage points · outputs/tables/contributions_monthly.csv · 6

FINDING 3 · SURGE VS COOLING

Transport led the surge, Housing leads the cooling

Category	Surge avg	Rank	Cooling avg	Rank
Transport	2.11 pp	1	0.37 pp	3
Food	1.19 pp	2	0.40 pp	2
Housing & Utilities	1.17 pp	3	0.49 pp	1
Recreation, Sport & Culture	0.39 pp	4	0.08 pp	5
Health	0.21 pp	5	0.34 pp	4

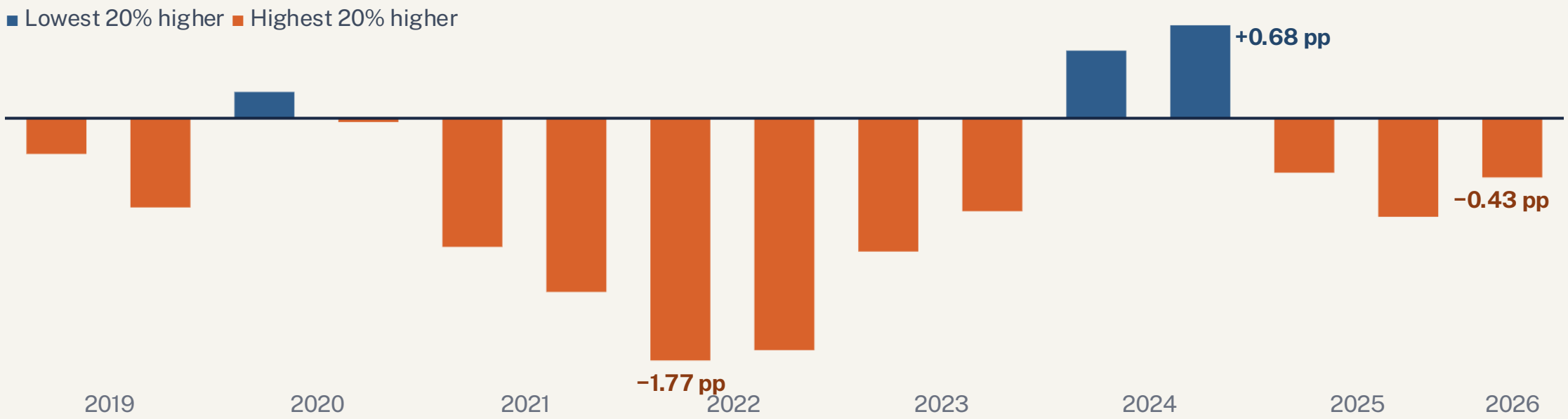
−83%

Transport's average monthly contribution fell from 2.11 pp to 0.37 pp, so Housing & Utilities moved to first place.

Surge = Jan 2022–Dec 2023; cooling = Jan 2024–Aug 2026 · average monthly contribution · outputs/tables/surge_vs_cooling.csv · 7

FINDING 4 · RICH VS POOR

The richest 20% usually faced higher inflation

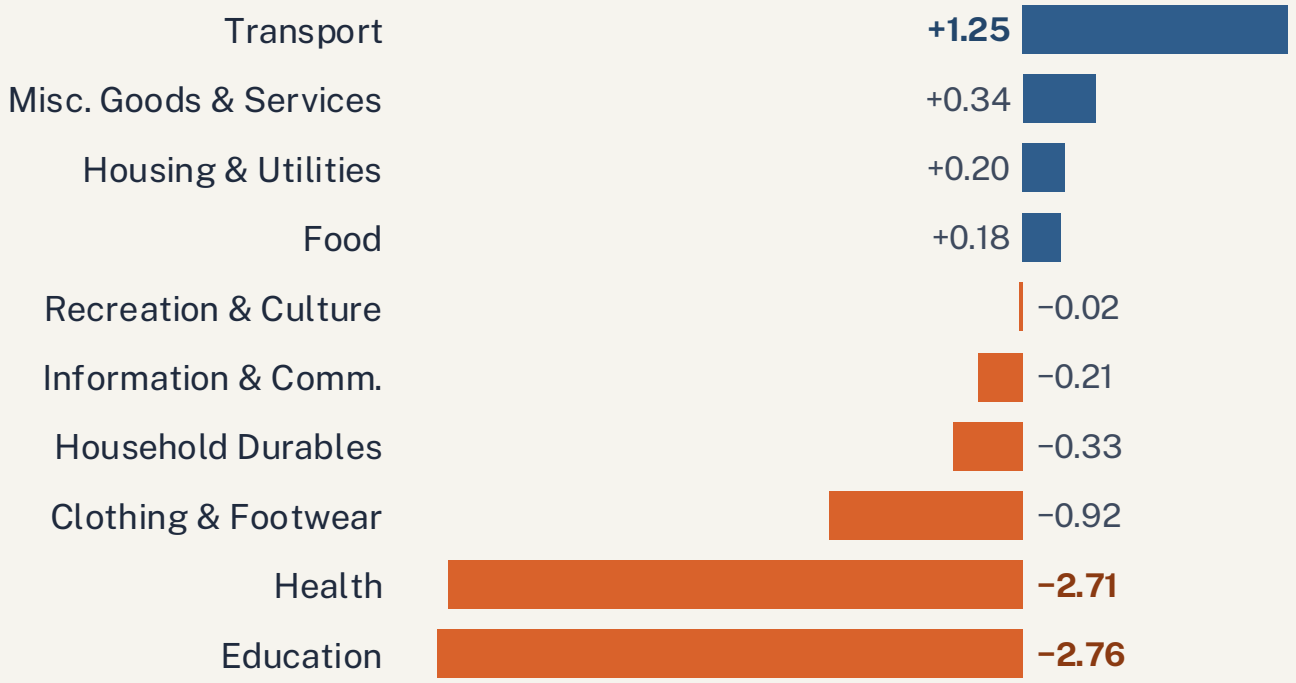


In 2026 1H: lowest 20% at **1.20%**, middle 60% at **1.75%**, highest 20% at **1.63%**. Only in 2024 did the lowest 20% face higher inflation.

Low-minus-high gap in year-on-year inflation, pp, two bars per year (1H, 2H) · outputs/tables/income_group_inflation_gaps.csv · 8

FINDING 5 · WHAT EXPLAINS THE GAP

Education and Health widen it, Transport narrows it



Health: prices in the richest group's basket rose 4.40%, vs 1.70% for the lowest 20%.

Education: fell 3.59% for the lowest 20%, but only 0.82% for the richest.

Transport: rose 5.87% for the lowest 20%, vs 4.62% for the richest.

2026 1H, low-minus-high gap in category inflation rates, pp · outputs/tables/income_gap_drivers.csv · 9

FINDING 6 · BEHIND THE NUMBERS

COE premiums and HDB rents rose faster than CPI

Private transport

292
COE premium index

150
Private Transport CPI

Housing

160
HDB median rent index

115
Accommodation CPI

Both pairs moved in the same direction since February 2021, but the CPI components rose far less. This is co-movement only, not a causal link.

Indexes, Feb 2021 = 100, value in Aug 2026 · LTA, HDB, SingStat · outputs/tables/transport_vs_coe.csv, accommodation_vs_hdb_rent.csv · 10

ROBUSTNESS

The decomposition holds up

97.8%

of months: contributions add up to headline inflation within 0.3 pp

0.39 pp

largest residual (Oct 2024); gaps appear only in 2019 and 2024, where the basket changes

Same top 3

under 2019 and 2024 weights from April 2026 onward

So we are confident about the latest-month ranking, but we do not claim one category was top throughout 2025–2026.

DASHBOARD DEMO

“What is pushing up prices, and does it hit my household harder?”

1
Overview & Category Explorer
Headline rate and any category over time

2
What Drove Inflation?
Contributions, surge vs cooling

3
Cumulative Price Changes
Transport up 51.7% since Jan 2019

4
Rich vs Poor
Income-group rates and category gaps

5
Behind the Numbers
Private Transport vs COE, rents vs CPI

Live: ds608-sg-inflation.streamlit.app

TAKEAWAYS

1. Three categories, Transport, Food and Housing & Utilities, explain most of Singapore's inflation since 2019.

2. Transport led the surge; Housing & Utilities leads the cooling.

3. By the price index, the richest 20% usually faced higher inflation, mainly through Health and Education.

LIMITATIONS

What this analysis cannot tell you

Descriptive, not causal

We show which categories contributed, not why prices changed.

Basket change in 2024

Contribution sizes are not comparable across 2019 and 2024 weights.

Coarse income data

Income-group CPI is half-yearly, with no income-group weights.

Average baskets

CPI tracks price changes, not price levels or any one household.

Small residual

Published weights and rounded indexes leave up to 0.39 pp unexplained.

Supporting data gaps

No COE for Apr–Jun 2020; HDB rents are owner-declared.

CONTRIBUTIONS

Who did what

DATA

Su Liang-Yu

Data collection, cleaning (wrangle.py), data documentation, repo setup

ANALYSIS

Jiang Liu Ru

Contribution decomposition, surge vs cooling, income gaps, robustness checks (analysis.py)

DASHBOARD

Amsyar Minhaj

Streamlit and Plotly dashboard, interactive controls, deployment

Thank you. Questions?